Neftel 2019 GBM — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

10x leaf scoped (SmartSeq2 non-integrable: platform iLISI~1.0 across θ=2–20, disjoint samples, UMI vs TPM). 7/10 samples (≥413 cells) on one connected component; sample iLISI 2.9/7.

Gate 2 — module geography → GO (non-branching)

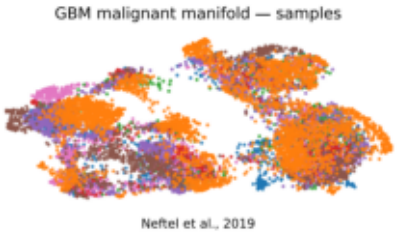
Shared Gast/pSom gradient (r=+0.79, co-vary) with Crest an inverse terminal readout (r=−0.89/−0.90). Curve fold-free (0 forks); Crest-high decile at pseudotime 0.76.

Gate 3 — cross-repression → NA

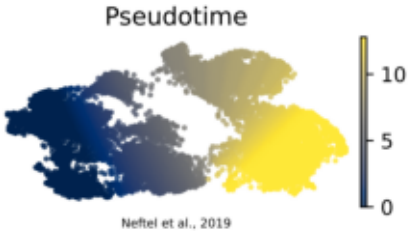
Gast/pSom co-vary (not compete) → single collapsed axis, no bifurcation to test.

STEP FIGURES

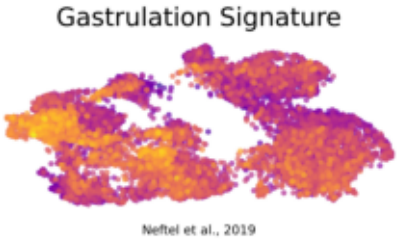
Manifold — samples



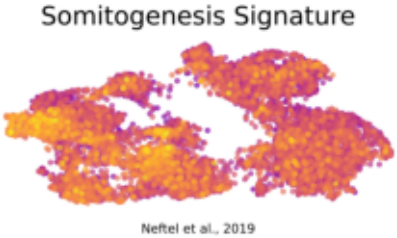
Pseudotime



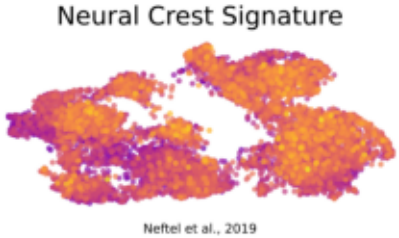
Gastrulation



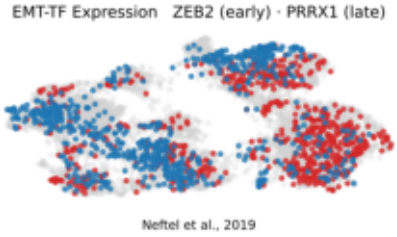
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Neftel et al., 2019); SmartSeq2+10x atlas, 10x leaf analyzed; 11,172 malignant raw → 10,229 final. Platform-exclusive patients make the SmartSeq2 leaf non-integrable with 10x.

Verdict.
Pattern reproduces as a non-branching embryonic axis: Gast/pSom-high origin → Crest-high terminus; developmental order pSom ρ=−0.74, Crest ρ=+0.55 along pseudotime, checked off a diffusion embedding independent of the fitted UMAP. Linear-pSom/Crest (pSom globally strongest).

Parameter choices

Min cells/sample	≥413 (mult-gap 647→264; 7/10)	**Integration**	Harmony θ=10; PCA20; spread=1.3
Scope	10x leaf only (SmartSeq2 non-integrable)	**Trajectory**	scFates curve N=20, λ=0.3, μ=100
Root node	Gast/pSom-high tip	**Geometry**	Linear (non-branching); 0 forks
EMT-TFs	ZEB2 (early) → PRRX1 (late)		